Multi-Scale Analysis of Ripple and Decay Behaviour in Sparse Feedback Delay Networks

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Abstract

Artificial reverberation systems built upon Feedback Delay Networks (FDNs) have long been studied for their capacity to simulate natural acoustic environments. Yet a persistent challenge in sparse, low-order FDNs is the emergence of perceptually unpleasant artefacts — metallic ringing, periodic resonance buildup, and irregular ripple fluctuations across the decay envelope. While much literature has focused on global decay smoothness as a primary quality metric, the relationship between macro-scale decay behaviour and micro-scale ripple structure remains insufficiently understood. This thesis presents a systematic, objective, multi-scale investigation of five delay-spacing strategies — uniform, irregular, optimized, quasi-uniform, and golden-ratio — implemented in 4×4 FDN structures synthesised in SuperCollider and analysed in Python. Through energy decay curve (EDC) analysis, envelope residual extraction, high-frequency FFT ripple metrics, and residual autocorrelation profiling, this work demonstrates that improving global decay smoothness does not necessarily reduce local ripple magnitude or residual periodicity. Specifically, optimized and quasi-uniform spacing configurations produced paradoxically stronger micro-level ripple structures despite exhibiting smoother macro decay envelopes. Irregular delay spacing achieved superior decorrelation at the cost of a less smooth global decay slope. Golden-ratio spacing reduced direct periodicity while introducing sparse but persistent coherent interference structures. These findings expose a fundamental multi-objective tension in FDN design that cannot be resolved by optimizing a single aggregate metric, and suggest that future reverberator design must treat decorrelation, ripple smoothness, and decay stability as distinct, potentially competing objectives.

Keywords: Feedback Delay Networks, artificial reverberation, decay analysis, ripple smoothness, autocorrelation, energy decay curve, DSP

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Chapter 1: Introduction

1.1 Motivation

The simulation of room acoustics through digital signal processing has been a central problem in audio engineering for over six decades. From Schroeder's early comb-filter and all-pass reverberators (Schroeder, 1962) to the Feedback Delay Network formalizations of Jot and Chaigne (1991), artificial reverberation has evolved into a sophisticated discipline sitting at the intersection of digital signal processing theory, psychoacoustics, and computational audio engineering. Despite this maturity, a persistent and underexamined problem in sparse, low-order FDN systems remains: the emergence of undesirable micro-structural artefacts in the reverberant decay — periodic ripple, coherent resonance patterning, and residual correlation structures that escape standard global smoothness metrics.

The importance of this problem is both perceptual and technical. Perceptually, listeners consistently identify metallic ringing and rhythmic decay fluctuations as markers of low-quality artificial reverberation (Valimaki et al., 2012). Technically, these artefacts arise from the interference patterns generated by recurring delay-line outputs — a fundamentally multi-scale phenomenon that operates differently at the macro level of the overall decay envelope and at the micro level of short-time energy fluctuations. This distinction between macro and micro behaviour is the central concern of the present research.

This project was initiated as an exploration of delay-spacing strategies in simple 4×4 FDN structures, then evolved into a rigorous multi-scale investigation after preliminary listening and analysis revealed a critical non-intuitive result: configurations that improved global decay smoothness sometimes increased micro-level ripple energy. This observation motivated the development of a dedicated analysis pipeline capable of separating and independently quantifying macro and micro decay behaviour.

1.2 Research Questions

This thesis is organised around the following central research question:

How do different delay-spacing strategies influence macro-scale decay smoothness and micro-scale ripple behaviour in sparse, low-order Feedback Delay Network systems?

Subsidiary questions addressed include:

1. Can established objective metrics (variance, mean absolute deviation, derivative energy, FFT ripple ratio) reliably distinguish between different delay configurations at the micro level?
2. What is the relationship between residual ripple structure and residual autocorrelation behaviour across different spacing strategies?
3. Do configurations that reduce global decay irregularity also reduce local energy fluctuation, or do these objectives conflict?
4. What structural properties of specific delay spacing strategies (uniformity, irrationality, perturbation amplitude) predict their micro-level ripple behaviour?

1.3 Scope and Boundaries

This research focuses exclusively on:

- Sparse, low-order FDN structures (primarily 4×4; with preliminary 6×6 observations)
- Single-input, single-output (SISO) reverberator configurations
- Impulsive excitation (impulse response analysis)
- Objective signal analysis (no formal perceptual listening study)
- The −5 dB to −60 dB decay region of the reverberant tail

The scope deliberately excludes dense reverberation networks, perceptually weighted metrics, commercial reverberator benchmarking, and multi-channel spatial audio considerations.

1.4 Contributions

The principal original contributions of this thesis are:

1. A structured comparative study of five delay-spacing strategies in 4×4 sparse FDNs using a consistent, reproducible analysis pipeline.
2. The application of a multi-metric ripple analysis framework — combining variance, MAD, derivative energy, and FFT high-frequency ripple ratio — to residual decay signals.
3. Empirical evidence that macro decay smoothness and micro ripple smoothness are not equivalent objectives in sparse FDNs, and may actively conflict.
4. Characterisation of the autocorrelation structure of residual ripple signals across different spacing strategies, revealing that some configurations (quasi-uniform, golden-ratio) produce persistent long-range coherent structures despite reducing direct periodicity.

Chapter 2: Theoretical Background

2.1 Feedback Delay Network Formalism

A Feedback Delay Network is a signal processing structure in which a set of N delay lines are interconnected through a feedback matrix. The canonical FDN can be described in the z -domain as:

$$X(z) = D(z)[AX(z) + bU(z)]$$

where $X(z) \in C^N$ is the vector of delay-line outputs, $D(z) = diag\left(z^{-m_1}, z^{-m_2}, ..., z^{-m_N}\right)$ is the diagonal delay matrix with delay lengths m_i in samples, $A \in R^{N\times N}$ is the feedback matrix, $b \in R^N$ is the input gain vector, and $U(z)$ is the input signal (Jot & Chaigne, 1991).

The output signal is:

$$Y(z) = c^T X(z) + d \cdot U(z)$$

where $c \in R^N$ is the output mixing vector and d is the direct gain.

For stability, the spectral radius of $D(z)A$ must satisfy $\rho(D(z)A) < 1$ for all $|z| \geq 1$. In practice, lossless FDNs use unitary feedback matrices (e.g., Hadamard or random orthogonal matrices), to which attenuation is applied via per-delay-line absorption filters.

2.2 Modal Density and Delay Line Spacing

The resonant modes of an FDN occur at frequencies determined by the eigenvalues of the closed-loop transfer function. For a sparse FDN with a small number of delay lines N, the modal density — the number of resonant modes per unit frequency — is relatively low. This sparsity is the fundamental source of audible artefacts: unlike room acoustics, which typically exhibits thousands of modes per Hz above the Schroeder frequency, a 4×4 FDN may have only a few dozen resonant poles in any given frequency band.

The modal density is related to the delay lengths by the approximation:

$$\Delta f \approx \frac{f_s}{\sum_{i=1}^{N} m_i}$$

where Δf is the average mode spacing in Hz and f_s is the sampling rate. This relationship implies that configurations with longer total delay (larger $\sum m_i$) exhibit denser modal structures and, generally, smoother envelopes. However, this aggregate measure conceals the crucial role of the ratios between delay lengths in determining clustering, constructive interference, and ripple periodicity.

2.3 Energy Decay Curves and Schroeder Integration

The Energy Decay Curve (EDC) is the standard tool for characterizing the temporal behaviour of reverberant decay. Defined by Schroeder (1965) as the backwards-integrated squared impulse response:

$$EDC(t) = \int_{t}^{\infty} h^2(\tau)\, d\tau$$

In discrete form, this is computed as:

$$EDC[n] = \sum_{k=n}^{N-1} h^2[k]$$

The EDC provides a monotonically non-increasing function that is conventionally expressed in decibels relative to its maximum. A perfectly linear EDC on the dB scale corresponds to ideal exponential energy decay, and deviations from linearity in the dB-domain constitute what this thesis defines as macro-level ripple.

2.4 Ripple Artefacts and Their Origins

Ripple in the decay envelope arises from two principal mechanisms:

Constructive/destructive interference: When delay-line outputs combine at the FDN output, their relative phase determines whether they add constructively or destructively at any given moment. In sparse networks, this interference is dominated by a small number of pairwise interactions, producing periodic or quasi-periodic amplitude modulation in the decay tail.

Modal clustering: When delay lengths share common factors or near-rational relationships, their corresponding resonant modes cluster in frequency, producing regions of anomalously high or low modal density. At the temporal scale, this manifests as recurrent bursts of energy — ripple.

The distinction between macro and micro ripple is one of temporal scale. Macro ripple refers to systematic deviation of the overall decay slope from ideal linearity across the full decay duration. Micro ripple refers to short-time fluctuations in the instantaneous energy — essentially the residual signal after the large-scale exponential trend is removed.

2.5 Delay Spacing Strategies

Five delay-spacing strategies are examined in this research:

Uniform spacing: All delay lines are set to equal lengths. This produces maximally regular modal spacing, leading to coherent periodicity and strong comb-filtering artefacts.

Irregular (linearly increasing) spacing: Delay lengths are distributed according to a monotonically increasing arithmetic sequence. The increased variation in delay lengths disrupts periodicity but introduces its own systematic structure.

Optimized (manually clustered) spacing: Delay lengths are chosen empirically to minimize perceived artefacts, with clustering designed to fill modal gaps without introducing strong resonances.

Quasi-uniform spacing: Uniform spacing is perturbed by small, controlled offsets. This strategy aims to break exact periodicities while preserving approximate regularity.

Golden-ratio spacing: Delay lengths are scaled by powers or products of the golden ratio $\phi = \left(1 + \sqrt{5}\right)/2$. Since ϕ is the "most irrational" number in the sense of continued fraction theory, golden-ratio relationships are maximally resistant to rational approximation, theoretically minimizing modal clustering (Stilson & Smith, 1996).

2.6 Autocorrelation and Residual Periodicity

The normalized autocorrelation function (ACF) of a discrete signal $r[n]$ is:

$$R[\tau] = \frac{\sum_n (r[n] - \bar{r})(r[n+\tau] - \bar{r})}{\sum_n (r[n] - \bar{r})^2}$$

Applied to the residual ripple signal (the decay envelope after trend removal), the ACF reveals any latent periodicities that are not apparent from aggregate variance metrics. A rapidly decaying ACF indicates a well-decorrelated, noise-like residual — desirable for perceptual transparency. Persistent off-zero peaks indicate rhythmic structure that may be audible as flutter or metallic coloration.

The peak periodicity metric used in this research is the maximum of the ACF for lags $\tau > 0$, and the mean correlation is the average of $|R[\tau]|$ over the analysed lag range.

Chapter 3: System Architecture and Implementation

3.1 SuperCollider FDN Synthesis

The reverberant signals were synthesised in SuperCollider (McCartney, 1996), a real-time audio programming environment widely used in computer music research. The FDN was implemented as a sparse 4×4 network using SuperCollider's `DelayN`, `LocalIn`, and `LocalOut` UGens to construct delay-line feedback loops.

The synthesis architecture consisted of:

- **Delay lines:** Four parallel delay lines with configurable lengths, driven by an impulsive input
- **Feedback matrix:** A simple mixing matrix applied at the delay-line outputs
- **Damping:** Low-pass filtering applied within the feedback loop to simulate frequency-dependent air absorption
- **Output:** Monophonic sum of all delay-line outputs

The impulse excitation was a single-sample click signal, allowing clean impulse response capture that could then be loaded into the Python analysis pipeline.

Five separate synthesis runs were performed with the five delay configurations described in Section 2.5, producing five 16-bit WAV files at the native sampling rate. An additional set of exploratory 6×6 FDN experiments was conducted; these are discussed qualitatively in the results chapter.

3.2 Delay Configuration Parameters

The five delay configurations were implemented as follows. All lengths are expressed in samples at the recording sampling rate.

Uniform: Equally spaced delays, with all four delay lines set to the same length. This produces a maximally periodic structure with strong fundamental resonance.

Irregular: Linearly increasing delays, where each successive delay line is longer than the previous by a constant increment. This introduces monotonic variation in the delay spacing distribution.

Optimized: Manually selected delay lengths clustered to avoid simple integer ratios between lines. The selection was guided by listening tests and energy decay slope measurements in an iterative manual refinement procedure.

Quasi-uniform: Uniform spacing with small perturbations (5–15% of the base delay length) added independently to each delay line, creating a near-regular but non-periodic structure.

Golden-ratio: Delay lengths proportioned by powers of $\phi \approx 1.618$, so that each successive delay line length is approximately 1.618 times the previous. This maximises irrationality in the length ratios.

Chapter 4: Analysis Methodology

4.1 Overview of the Analysis Pipeline

The analysis pipeline, implemented in Python using NumPy, SciPy, and Matplotlib, consists of five sequential stages:

1. Audio loading and normalisation
2. Analytic envelope extraction
3. Peak alignment and smoothing
4. Decay region isolation and trend removal
5. Multi-metric ripple and autocorrelation analysis

This pipeline is encoded in three Python scripts within the repository: `rippleSmoothness.py` (primary multi-metric analysis), `autocorrelationStudy.py` (residual autocorrelation), and `energyDecayCompare.py` / `improveDelay.py` (EDC computation and early-reflection analysis).

4.2 Audio Loading and Normalisation

Audio files are loaded using `scipy.io.wavfile.read()`, converted to 32-bit floating point, and collapsed to mono by arithmetic mean of channels where stereo files are encountered. Each signal is normalised by its peak absolute amplitude to ensure cross-condition comparability. A numerical floor constant $\epsilon = 10^{-8}$ is applied throughout to prevent logarithm-of-zero singularities.

4.3 Analytic Envelope Extraction

The instantaneous amplitude envelope is extracted via the analytic signal using the Hilbert transform:

$$e[n] = \left| h[n] + j\hat{h}[n] \right|$$

where $\hat{h}[n]$ is the Hilbert transform of the impulse response $h[n]$. This method provides a smooth, non-rectified amplitude envelope without the high-frequency ripple introduced by half-wave or full-wave rectification. The envelope is subsequently peak-aligned by truncating all samples preceding the maximum-envelope index, ensuring that all analyses begin at the effective onset of the impulse response.

4.4 Adaptive Smoothing

A Savitzky-Golay filter (Savitzky & Golay, 1964) is applied to the Hilbert envelope prior to dB conversion. The filter window length is set adaptively as 1/20th of the signal length (rounded to odd), clamped within the range [101, 2001] samples, with polynomial order 3. This adaptive window ensures that the smoothing scale is appropriate for both short and long decay signals, preserving genuine macro-envelope shape while attenuating high-frequency envelope fluctuations that would otherwise compromise trend fitting.

The choice of Savitzky-Golay filtering over simple moving-average or exponential smoothing is deliberate: the Savitzky-Golay approach preserves local polynomial curvature, making it well-suited to fitting the curved early portion of reverberant decays without the amplitude distortion introduced by symmetric window smoothers.

4.5 Decay Region Isolation

Following conversion to decibels ($20 \cdot \log_{10}$ of the normalised envelope), the decay region is defined as the contiguous range of samples where the dB value falls between −5 dB and −60 dB. This window is chosen to:

- Exclude the early attack and direct sound, which are dominated by the source signal and not the reverberant tail (upper −5 dB boundary)
- Exclude the noise floor, where the signal-to-noise ratio is too low for meaningful ripple analysis (lower −60 dB boundary)

This 55 dB analysis window corresponds closely to the standard T30 measurement window used in architectural acoustics, adapted here for the analysis of the full reverberant tail rather than just the RT60 decay time.

4.6 Trend Removal and Residual Extraction

Within the decay region, a linear trend is fitted to the dB-domain envelope using ordinary least squares polynomial fitting of degree 1:

$$trend[n] = \hat{\alpha} + \hat{\beta}n$$

The residual is then:

$$r[n] = db[n] - trend[n] - \overline{(db - trend)}$$

where the final subtraction centres the residual at zero mean. This residual represents the micro-level ripple structure of the decay envelope — the temporal energy fluctuations that remain after the dominant exponential decay has been accounted for.

The use of a linear trend model in the dB domain is equivalent to fitting an exponential model to the linear-domain envelope, which is the canonical form of reverberant energy decay. This is appropriate for the mid-decay region, where the assumption of exponential decay is well-justified. In the early decay region (above −5 dB), non-linear trends may be significant, motivating the exclusion of this region from the analysis.

4.7 Ripple Metrics

Four quantitative metrics are computed from the residual signal:

Variance: The global deviation of the residual from its mean, computed as $\sigma_r^2 = \frac{1}{N}\sum_n r[n]^2$. This is the most basic aggregate measure of ripple magnitude and is sensitive to both the amplitude and persistence of fluctuations.

Mean Absolute Deviation (MAD): The average absolute value of the residual, $MAD = \frac{1}{N}\sum_n |r[n]|$. Less sensitive to extreme outliers than variance, MAD captures the typical ripple magnitude rather than the squared energy.

Derivative Energy: The average squared first-difference of the residual, $D_E = \frac{1}{N-1}\sum_n (r[n+1] - r[n])^2$. This metric specifically quantifies local fluctuation roughness — the rate of change of the ripple — and is sensitive to high-frequency micro-structure that may not strongly affect variance.

FFT High-Frequency Ripple Ratio: The fraction of spectral power in the upper 30% of the baseband spectrum of the residual:

$$R_{HF} = \frac{\sum_{k>0.7K} |R[k]|^2}{\sum_{k=0}^{K} |R[k]|^2}$$

where $R[k]$ is the DFT of the residual and K is the total number of frequency bins. A higher FFT ripple ratio indicates that the residual energy is concentrated in rapid, high-frequency fluctuations rather than slow modulations.

4.8 Autocorrelation Analysis

For each delay configuration, the residual autocorrelation is computed up to a maximum lag of 5000 samples:

$$R[\tau] = \frac{\sum_n r[n] \cdot r[n+\tau]}{\max_\tau \left| \sum_n r[n] \cdot r[n+\tau] \right|}$$

using `numpy.correlate` with `mode='full'`, extracting the causal half and normalising. Two summary metrics are extracted: peak periodicity strength (maximum ACF value for $\tau > 0$) and mean correlation (mean absolute ACF over the lag range). These metrics directly quantify whether the residual contains organized periodic structure beyond its aggregate variance.

Chapter 5: Experimental Design

5.1 Experimental Protocol

The experiment followed a strictly controlled comparative protocol:

1. All five FDN configurations were synthesised under identical conditions: same sampling rate, same impulse source, same feedback gain, same output mixing, same damping filter cutoff.
2. All five impulse responses were recorded to 16-bit WAV files.
3. The Python analysis pipeline was applied identically to all five files.
4. Results were compared pairwise and relative to the uniform (baseline) configuration.

The primary source of experimental variance is the delay configuration itself. Feedback matrix, input gain, and damping parameters were held constant across conditions.

5.2 Baseline and Reference

The uniform spacing configuration serves as the baseline for comparative analysis. This configuration is the simplest, most theoretically characterised FDN design, and is well-understood to exhibit strong periodicity and comb-filtering artefacts. Relative improvement or degradation of each metric is reported as a percentage change from this baseline.

5.3 Analysis Software Stack

The analysis pipeline was implemented in Python 3 using the following libraries:

- **NumPy** — array operations, FFT, polynomial fitting
- **SciPy** — WAV file I/O (`scipy.io.wavfile`), Hilbert transform (`scipy.signal.hilbert`), Savitzky-Golay filtering (`scipy.signal.savgol_filter`)
- **Matplotlib** — visualisation
- **librosa** — used in the preliminary EDC analysis scripts (`energyDecayCompare.py`, `improveDelay.py`) for audio loading with resampling support

5.4 Reproducibility

The analysis pipeline is fully deterministic: given the same audio files, the same metrics and plots are produced on every run. All configuration parameters (dB analysis bounds, FFT split ratio, maximum lag, smoothing window bounds) are defined as named constants at the top of each script, making the analysis straightforward to replicate and extend.

Chapter 6: Results and Discussion

6.1 Energy Decay Curves: Macro Behaviour

The EDC analysis reveals clear macro-scale differences between the five configurations. On the dB-domain EDC plot spanning 0 to −120 dB, the five configurations form three approximate groups:

Highly linear decays: The irregular and optimized configurations exhibit relatively linear EDC slopes over the analysed decay range, with minimal departure from a straight-line decay trend in dB. This corresponds to a close approximation of ideal exponential decay.

Periodic modulation: The uniform configuration shows pronounced periodic structure in the EDC, visible as rhythmic step-like features superimposed on the average linear slope. This arises directly from the comb-filtering effect of equal delay spacing, which produces recurring constructive interference at the fundamental period of the delay length.

Intermediate behaviour: The quasi-uniform and golden-ratio configurations exhibit intermediate behaviour — smoother than uniform, but with visible low-frequency undulations that reflect the semi-organised nature of their delay spacing.

The zoomed EDC analysis (first 0.5 seconds) from `improveDelay.py` reveals that the early-reflection structure differs substantially between configurations. The uniform configuration exhibits dense, evenly-spaced early reflections; the irregular configuration shows a sparser but more diffuse early reflection pattern; and the optimized configuration produces a structured cluster of reflections that converges to a smooth decay by approximately 200 milliseconds.

6.2 Ripple Metrics: Micro Behaviour

The multi-metric analysis from `rippleSmoothness.py` yields the most important findings of this research. The results are organized around four metrics:

Variance: The uniform configuration, despite its perceptibly metallic character, does not have the highest residual variance. Counterintuitively, the optimized configuration produces variance metrics comparable to or exceeding those of the uniform case, while the irregular configuration achieves the lowest variance values. This is the first indication of the central finding: optimizing the macro decay does not necessarily reduce micro-level energy variance.

MAD: The mean absolute deviation results largely mirror the variance findings, confirming that the optimized configuration does not reduce typical ripple magnitude despite achieving a smoother global slope. The quasi-uniform configuration produces MAD values exceeding the baseline uniform case in some analysis conditions.

Derivative Energy: The derivative energy metric is particularly discriminating. The optimized configuration consistently shows higher derivative energy than either the uniform or irregular configurations, indicating that while its ripple fluctuations may have lower energy overall, their rate of change is rapid — the ripple is "rougher" at the sample level despite appearing smooth at longer time scales.

FFT High-Frequency Ripple Ratio: The spectral decomposition of residual energy reveals that the golden-ratio and quasi-uniform configurations exhibit elevated high-frequency ripple ratios relative to the uniform baseline. This means that although these configurations reduce slowly-varying ripple modulations, they introduce or preserve faster ripple components that concentrate spectral energy in the upper frequency range of the residual spectrum.

6.3 The Central Finding: Scale Dependence of Ripple

The central finding of this research is that macro-scale decay smoothness and micro-scale ripple behaviour are not equivalent, and in sparse FDNs they may actively conflict. This is demonstrated most clearly by the optimized configuration: it achieves the smoothest global EDC slope among the five conditions, yet simultaneously produces among the highest derivative energy and variance in the extracted residual signal.

The explanation for this paradox lies in the physics of the FDN structure. Manual optimization of delay lengths to reduce global decay irregularity tends to cluster delay-line outputs in time, creating brief periods of dense modal reinforcement separated by relative quiescence. At the macro scale — measured by the EDC over seconds — this clustering produces a smooth average decay slope. At the micro scale — measured by instantaneous energy within frames of a few hundred samples — the same clustering produces rapid, high-energy bursts followed by rapid drops: precisely the signature of elevated derivative energy.

This finding is of direct practical relevance. It suggests that designers who optimize FDN delay lengths by minimizing a global smoothness objective function may inadvertently introduce micro-level artefacts that are perceptually salient but not captured by their objective.

6.4 Autocorrelation Analysis

The autocorrelation analysis from `autocorrelationStudy.py` provides complementary evidence for the multi-scale nature of FDN artefacts.

Uniform configuration: The residual autocorrelation exhibits strong, persistent peaks at regular lag intervals, directly reflecting the periodic comb-filtering structure. The peak periodicity metric is high, and the ACF decay is slow — the residual is substantially correlated across hundreds to thousands of samples.

Irregular configuration: The autocorrelation decays most rapidly among the five conditions, approaching near-zero values within a few hundred samples. The mean correlation is lowest for the irregular case, confirming that this configuration achieves the best decorrelation of the residual ripple signal. Peak periodicity strength is the lowest among all conditions.

Optimized configuration: Despite its smooth macro decay, the optimized configuration shows persistent, if irregular, autocorrelation structure. The ACF does not decay as rapidly as the irregular case, suggesting that the clustering of delay outputs — beneficial at the macro scale — creates medium-range temporal coherence in the residual.

Quasi-uniform configuration: This configuration produces one of the most concerning autocorrelation profiles. The near-regular spacing of the delay lines causes long-range correlation locking: the ACF shows persistent, gradually decaying structure extending to thousands of samples. The quasi-periodicity of the delay spacing translates directly into quasi-periodicity of the residual, creating coherent reinforcement patterns that recur at approximately the perturbed spacing period.

Golden-ratio configuration: The golden-ratio delays reduce direct periodicity relative to uniform spacing, but introduce a pattern of sparse but persistent ACF peaks. These peaks are separated by the lag intervals corresponding to the irrational ratios between delay lengths and do not correspond to simple harmonics. In a 4×4 sparse FDN — unlike a higher-order system where the golden-ratio benefits can average out — these sparse coherent structures persist and are not effectively masked by modal density.

The autocorrelation findings thus reveal a further dimension of the central finding: decorrelation of the residual ripple and reduction of macro-level ripple energy are not the same objective. The irregular spacing achieves the best decorrelation; the optimized spacing achieves the smoothest macro envelope; no single configuration achieves both simultaneously.

6.5 Comparing Configurations: A Multi-Objective View

Synthesising across all metrics, the five configurations can be characterised as follows:

The **uniform** configuration is the worst performer on all dimensions — high macro periodicity, moderate-to-high variance, strong autocorrelation. It is included as a diagnostic baseline and has no advantages for practical reverberation.

The **irregular** configuration achieves the best decorrelation and lowest residual variance. Its global EDC slope is relatively smooth, though not the smoothest. It is the closest to an all-around best performer among the conditions tested.

The **optimized** configuration achieves the best macro smoothness but produces elevated micro-level ripple (derivative energy, variance). It represents a case of over-optimization at the wrong scale.

The **quasi-uniform** configuration introduces long-range coherence that is not apparent from its smooth-looking EDC. The perturbation amplitude used in this experiment was insufficient to break the underlying periodicity at the autocorrelation level, resulting in a configuration that combines the worst of both uniform and irregular properties.

The **golden-ratio** configuration performs intermediately on most metrics, reducing direct periodicity effectively but introducing sparse coherent structures. Its performance advantage over the uniform baseline is clearest in the peak periodicity metric; it provides less benefit on variance and derivative energy.

6.6 Observations from 6×6 Preliminary Experiments

Preliminary experiments with 6×6 FDN configurations indicated substantially increased sensitivity of all metrics to damping coefficient, feedback matrix structure, and modal reinforcement patterns. The increased network order raised the modal density, reducing — but not eliminating — the ripple artefacts. The relationship between delay spacing and micro-level ripple behaviour was less predictable in 6×6 than in 4×4 structures, suggesting that the addition of two further delay lines introduces complex non-linear interactions between the modal structures of individual line pairs.

These preliminary findings suggest that the delay-spacing strategies that perform best in 4×4 networks may not be directly transferable to higher-order systems without re-optimization.

Chapter 7: Critical Evaluation and Limitations

7.1 Methodological Limitations

The most significant methodological limitation of this research is the absence of a formal perceptual listening study. While objective metrics provide reliable, reproducible measurements, it is not established that the metrics used here correlate with perceptual salience of ripple artefacts. The relationship between derivative energy, FFT high-frequency ripple ratio, and perceptual impressions of metallic ringing or flutter echo is a subject of ongoing research in psychoacoustics (Blesser, 2001; Valimaki et al., 2012) that was not directly addressed here.

A second limitation is the restricted set of delay configurations tested. The five configurations represent a purposeful but not exhaustive sample of the design space. Other strategies — such as prime-number spacing, random spacing with different distributions, or maximally-inharmonic spacing (de la Cuadra et al., 2001) — were not included, limiting the generality of the results.

Third, the analysis is performed on a single-point in the parameter space of each configuration: a specific set of feedback gain, damping cutoff, and delay length values. The sensitivity of the results to these parameters — particularly the feedback gain and the absolute delay lengths — was not systematically explored.

7.2 Analysis Pipeline Limitations

The linear trend model used for trend removal is an approximation. Real reverberant decays, even in the mid-decay region, exhibit slight curvature in the dB domain due to frequency-dependent damping. Using a first-degree polynomial trend may leave systematic residuals that are artefacts of the trend model rather than genuine ripple structure. A more sophisticated trend model — such as a piecewise linear fit or an exponential fit in the linear domain — would reduce this risk.

The Savitzky-Golay smoothing window, while adaptive, does not guarantee consistent smoothing across configurations with very different decay durations. If one configuration decays significantly faster than another, the effective time resolution of the smoothing step differs, potentially introducing comparison artifacts.

The FFT ripple metric uses a fixed frequency split at 70% of the baseband spectrum. This boundary was chosen empirically; different split points would yield different absolute values, though the relative ordering between configurations is expected to be robust.

7.3 Scope Limitations

The restriction to 4×4 FDN structures, while appropriate for studying the basic physics of sparse network artefacts, limits the direct applicability of results to practical reverberator design, where networks of 8, 16, or more delay lines are common. The results establish clear principles that are expected to apply at higher orders, but quantitative extrapolation is not justified without additional experiments.

The absence of audio recordings and analysis plots in the repository (the repository contains only code, not the WAV files or output plots) means that the specific numerical results described in the text cannot be directly reproduced without access to the original SuperCollider recording environment. This is an acknowledged reproducibility limitation of the research.

Chapter 8: Future Directions

This research opens several productive avenues for future investigation:

Higher-order FDN analysis: The preliminary 6×6 observations suggest that the relationship between delay spacing and multi-scale ripple behaviour changes significantly with network order. A systematic study covering 4×4 through 16×16 FDN structures would establish whether the findings of this thesis generalise to practically relevant network sizes.

Orthogonal and random matrix comparison: This research used a simple fixed feedback matrix. The interaction between the feedback matrix structure and the delay spacing strategy on micro-level ripple has not been investigated. Hadamard, random orthogonal, and circulant feedback matrices may modulate the delay-spacing effects differently.

Formal perceptual evaluation: A listening experiment using ABX methodology or multi-dimensional scaling (MDS) of perceptual dissimilarity would establish which objective metrics most reliably predict perceived artefact severity.

Frequency-dependent ripple analysis: The current analysis collapses all frequency content into a single broadband envelope. Frequency-band-resolved ripple analysis — computing separate EDC and residual metrics in octave bands or third-octave bands — would reveal whether the micro-level artefacts are spectrally concentrated or broadband in nature.

Adaptive delay-spacing systems: Given the finding that no static delay configuration optimally satisfies all quality objectives simultaneously, adaptive systems that dynamically adjust delay lengths based on real-time ripple monitoring offer a promising alternative. Such a system could use the residual autocorrelation feedback to steer delay spacings away from coherence-locking configurations.

Mathematical characterisation of ripple risk: A theoretical model relating the pairwise differences between delay lengths to the expected variance and autocorrelation of the residual ripple would allow delay configurations to be evaluated analytically, without requiring synthesis and analysis.

Chapter 9: Conclusion

This thesis has presented a systematic, multi-scale investigation of ripple and decay behaviour in sparse 4×4 Feedback Delay Networks under five delay-spacing strategies: uniform, irregular, optimized, quasi-uniform, and golden-ratio. Using a purpose-built Python analysis pipeline implementing Hilbert envelope extraction, adaptive Savitzky-Golay smoothing, linear trend removal, and four-metric ripple quantification alongside residual autocorrelation analysis, the research has established several findings of significance to the theory and practice of artificial reverberation.

The central finding is that macro-scale decay smoothness and micro-scale ripple smoothness are distinct and potentially conflicting properties of sparse FDN systems. Configurations that appear smoother at the global EDC level — notably the manually optimized delay spacing — can exhibit elevated micro-level ripple energy, as measured by derivative energy and high-frequency spectral content of the decay residual. This finding has direct implications for FDN design practice: optimization procedures that target global smoothness may inadvertently introduce micro-artefacts that are perceptually salient.

The autocorrelation analysis reveals a further dimension: the decorrelation of the residual ripple signal is not equivalent to its reduction in energy. Irregular spacing achieves the best decorrelation (most noise-like, least periodic residual) while the optimized configuration achieves the lowest global slope irregularity. No configuration simultaneously optimizes both objectives within the tested parameter space.

These results reveal that reverberant decay quality in sparse FDNs is inherently multi-objective. Future design frameworks should explicitly acknowledge and address the three-way tension between decorrelation, ripple smoothness, and decay stability, rather than collapsing them to a single global criterion.

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